



Index Number:

NATIONAL CERTIFICATE OF EDUCATION

2024

PHYSICS (N530) TIME: 45 MINUTES

Candidates answer on the Question Paper.
Additional Materials: Mathematical set, Calculator

READ THESE INSTRUCTIONS FIRST

1. Write your index number in the space provided above.
2. Write in dark blue or black ink. Do not use correction fluid.
3. You may use a soft pencil for any diagram, graph or rough working.
4. Diagrams are not drawn to scale unless otherwise specified.
5. Any rough working should be done in this booklet.
6. Answer **ALL** questions.
7. This document consists of **6** questions printed on **12** pages, numbered **2** to **13**.
8. The total marks for each question is shown in brackets () at the beginning of each question.
9. The number of marks for each part question is shown in brackets [].
10. The total number of marks for this paper is **50**.

For Examiners' use								
Question No.	1	2	3	4	5	6	Total	Signature
Examiner								
Team Leader								
CE/ACE								

Question 1 (10 marks)

Circle the correct answer. Each item carries one mark.

(a) Which of the following is a unit of **temperature**?

- A** centimetre (cm)
- B** kelvin (K)
- C** gram (g)
- D** metre (m)

(b) A ruler is used to measure the length of an object as shown in Fig. 1.1.

In Fig. 1.1, circle the letter which shows the correct position of the eye to avoid parallax error.

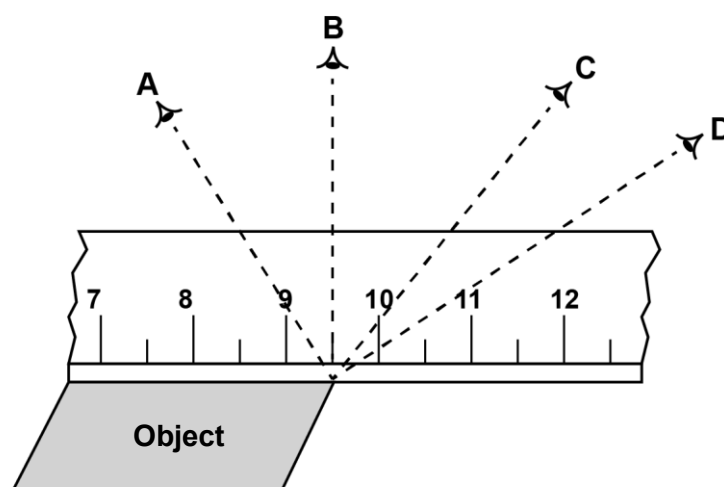


Fig. 1.1

- (c) To see in the dark, we switch on light bulbs.

This is because light bulbs

- A** absorb light.
- B** block light.
- C** emit light.
- D** reflect light.

- (d) A man runs a distance of 100 m in 20 s.

What is the average **speed** of the man?

- A** 200 m/s
- B** 120 m/s
- C** 80 m/s
- D** 5 m/s

- (e) Fig. 1.2 shows three beams of light labelled **X**, **Y** and **Z** respectively.

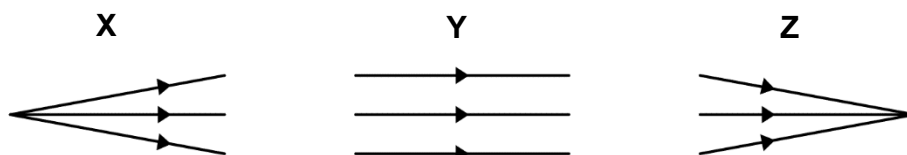


Fig. 1.2

Which of the following shows both the **parallel** and the **convergent** beams?

	Parallel beam	Convergent beam
A	Z	X
B	Y	Z
C	Y	X
D	X	Z

- (f) A student determines the volume of a stone using the set-up shown in Fig. 1.3 (a) and Fig. 1.3 (b).

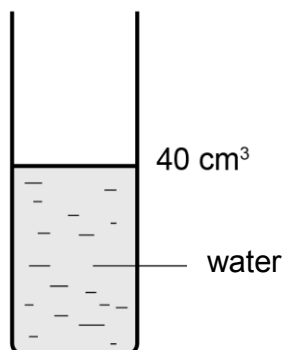


Fig. 1.3 (a)

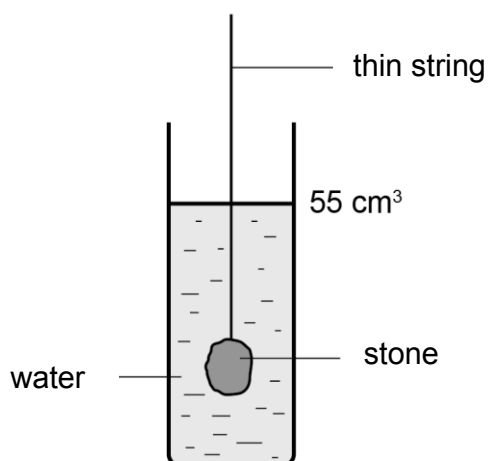


Fig. 1.3 (b)

What is the **volume of the stone**?

- A** 15 cm³ **B** 40 cm³
C 55 cm³ **D** 95 cm³
- (g) Fig 1.4 shows a stopwatch.

What is the reading shown on the stopwatch?

- A** 18 s
B 32 s
C 1 min 15 s
D 2 min 18 s

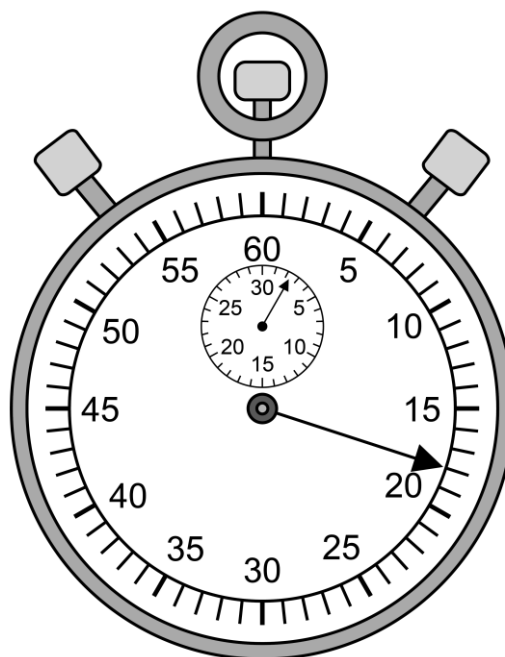


Fig 1.4

(h) What **form** of energy is stored in a dry cell?

- A** Chemical
- B** Heat
- C** Light
- D** Sound

(i) A boy walks from **P** to **Q** to **R** and back to **P** as shown in Fig. 1.5.

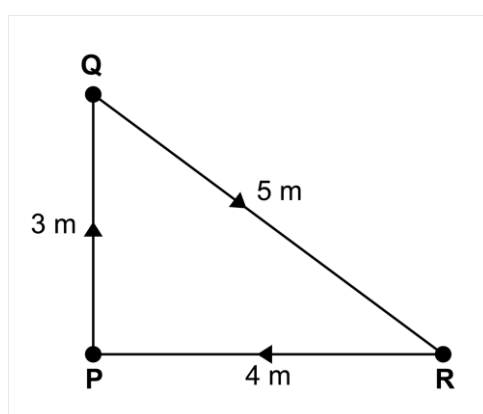


Fig. 1.5

What is his **total displacement**?

- | | |
|---------------|---------------|
| A 0 m | B 8 m |
| C 12 m | D 16 m |

(j) According to the law of conservation of energy, which of the following statements is **true**?

- A** Energy can be created.
- B** Energy can be destroyed.
- C** Energy can change from one form to another.
- D** Energy can either be renewable or non-renewable.

E
TL
CE

Question 2 (5 marks)

Fig. 2.1 shows a ray of light striking the surface of a plane mirror **MN**.

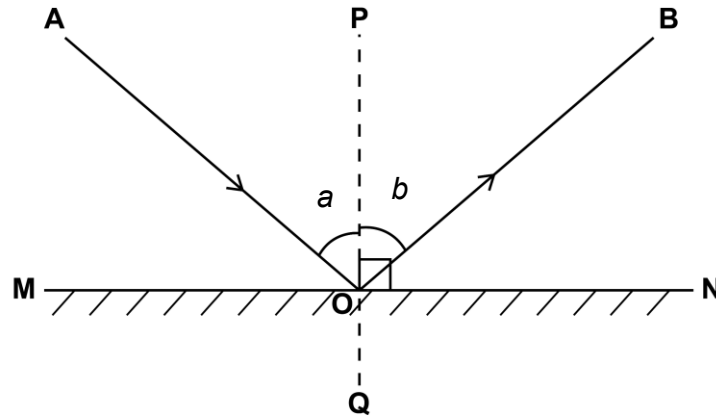


Fig. 2.1

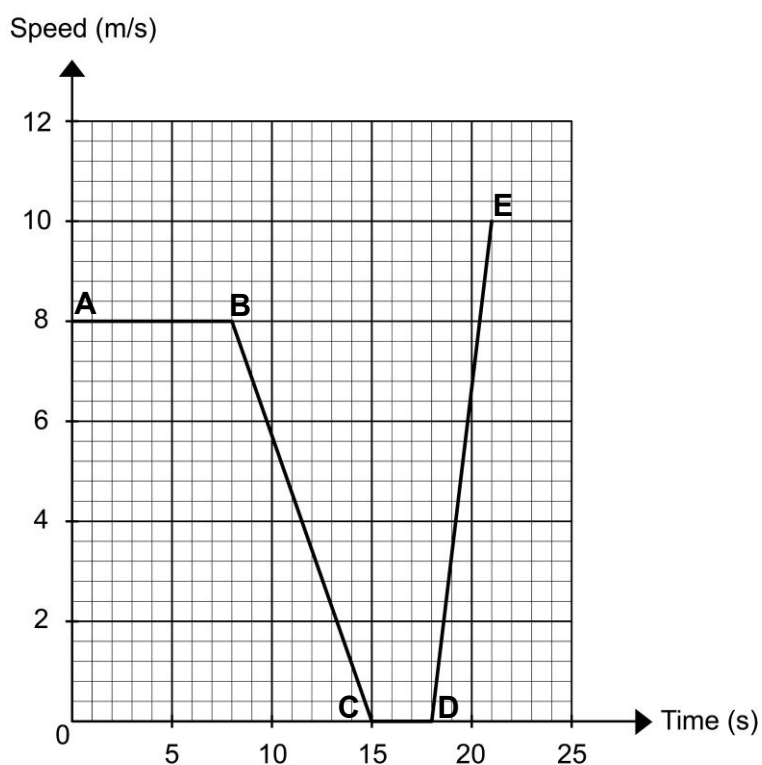
Match each item in **Column A** with its correct label in **Column B**.

Column A		Column B
(a) Reflected ray	•	• <i>b</i>
(b) Angle of incidence	•	• PQ
(c) Angle of reflection	•	• AO
(d) The normal at the point of incidence	•	• ON
(e) Incident ray	•	• <i>a</i>
		• OB

[5]

Question 3 (5 marks)

Fig. 3.1 shows the speed-time graph of a car moving along a straight line.

**Fig. 3.1**

- (a) (i) For how long does the car move at a uniform speed of 8 m/s?

..... [1]

- (ii) What is the **maximum** speed reached by the car on its journey?

..... [1]

- (b) Fully describe the motion of the car between

- (i) C and D.

..... [1]

- (ii) B and C.

.....

 [2]

E
TL
CE

Question 4 (10 marks)

- (a) Complete the following sentences with the correct word from the given list.

kinetic **light** **non-renewable** **potential**
renewable **chemical**

[**Note:** there is one extra word in the list]

- (i) In a thermal power station, energy stored in coal is converted into electrical energy.
- (ii) A ball found on the roof of a house has energy.
- (iii) Wind is a source of energy.
- (iv) A thermal power station uses the energy of steam to turn a turbine.
- (v) A disadvantage of using coal to produce electricity is that it is a source of energy.

[5]

- (b) (i) Which **two** quantities from the list below are vector quantities?

Tick (✓) the **two** correct boxes.

- | | |
|--------------------------|--------------|
| ☐ | Distance |
| ☐ | Displacement |
| ☐ | Speed |
| ☐ | Velocity |
| ☐ | Mass |

[2]

- (ii) A car moving along a straight line accelerates uniformly from a speed of 10 m/s to 20 m/s in 2 s.

Calculate the **acceleration** of the car.

Show all your workings.

Acceleration = m/s² [3]

E
TL
CE

Question 5 (5 marks)

A ball attached to the lower end of a string is set swinging freely between positions **A** and **C** as shown in Fig. 5.1.

When the ball is at position **B**, it is just above the ground.

When it is at positions **A** and **C**, it is at a height of 0.6 m above the ground.

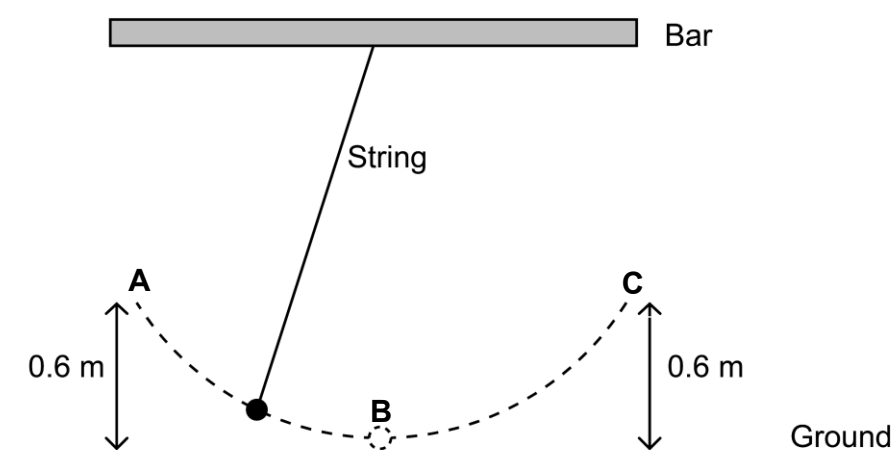


Fig. 5.1

- (a) The ball has a mass of 0.5 kg.

What is the gravitational potential energy of the ball at position **A**? [Take $g=10 \text{ m/s}^2$].

Show all your workings.

Gravitational potential energy = J

[2]

(b) Tick (✓) the correct box.

(i) When the ball moves from **A** to **B**, its **gravitational potential energy** decreases.

This is because

☐

the mass of the ball decreases.

☐

the speed of the ball decreases.

☐

the height of the ball decreases.

[1]

(ii) What happens to the **kinetic energy** of the ball when it moves from **A** to **B**?

Tick (✓) the correct box.

☐

It decreases.

☐

It increases.

☐

It is unchanged.

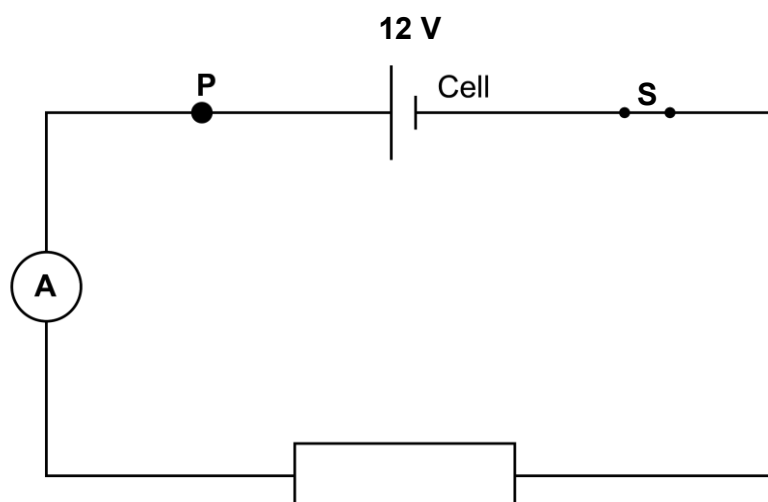
[1]

(c) Assuming there is no loss of energy, state the kinetic energy of the ball at position **B**.

Kinetic energy = J [1]

Question 6 (15 marks)

- (a) Fig. 6.1 shows an electric circuit in which a current is flowing.

**Fig. 6.1**

- (i) Indicate the positive terminal of the cell by drawing a + sign on Fig. 6.1. [1]
- (ii) In Fig 6.1, draw an arrow to show the direction of current at point **P**. [1]
- (iii) Give a reason why current flows in the circuit when the switch **S** is closed.
 [1]
- (iv) 600 C of charge flows in the circuit in 120 s.
 Calculate the current flowing in the circuit. Show all your workings.

Current = A [2]

- (v) The e.m.f. of the cell is 12 V.
 How much work is done to drive 600 C of charge around the whole circuit?

Work done = J [2]

- (b) Fig. 6.2 shows a **different** electric circuit.

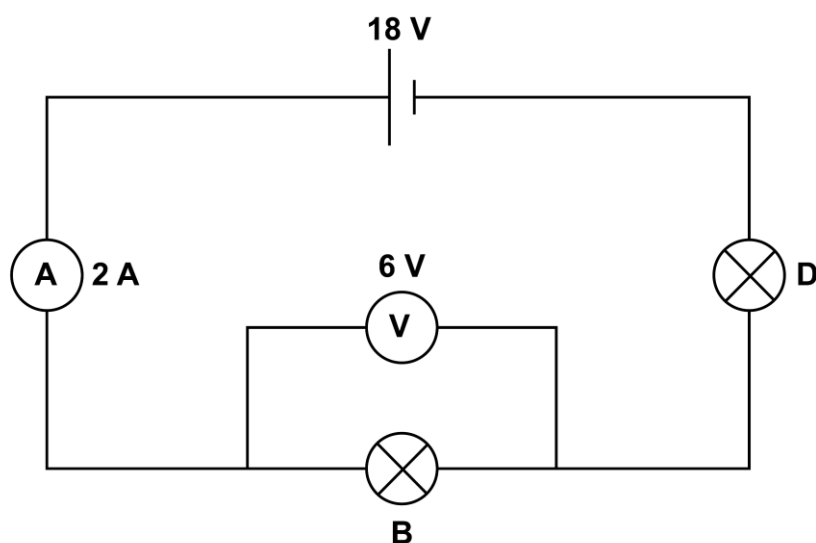


Fig. 6.2

- (i) The e.m.f. of the cell is 18 V. The voltmeter reading is 6 V.
What is the potential difference across bulb **D**? Show all your workings.

Potential difference = V [2]

- (ii) The ammeter reading is 2 A.
Calculate the **resistance of bulb D**. Show all your workings.

Resistance = Ω [2]

- (iii) Calculate the **total resistance** of the bulbs **B** and **D**.

Total resistance = Ω [2]

- (iv) A third bulb is connected in series with bulbs **B** and **D**.

Describe **one** change you observe.

..... [1]

- (v) Explain why this change happens.

..... [1]
.....

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